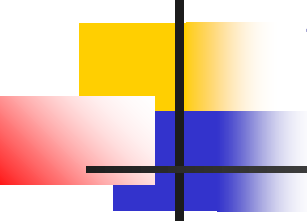


Deep Learning

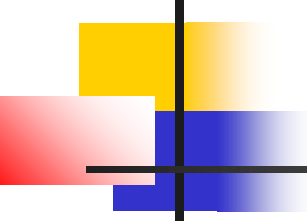
Kiến thức → Chia sẻ → Hiểu biết → Những ứng dụng có ích cho cộng đồng

Dr. Nguyễn Thị Kim Ngân



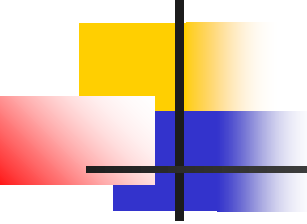
Prerequisites

- Knowledge: Calculus, Linear Algebra, Python Programming



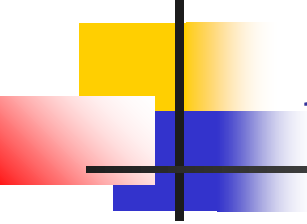
Course Objectives

- Introduce key deep learning methods
- Apply these methods to real-world problems



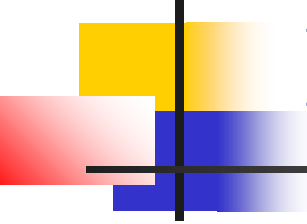
Course Content

1. Introduction to Deep Learning
2. Linear Regression
3. Logistic Regression
4. Neural Network
5. Convolutional Neural Networks
6. Object Detection
7. Recurrent Neural Networks
8. Sequence to sequence



Assessment

- Classwork: 50%
 - Attendance (10%)
 - Midterm Exam (40%): Exercises, in-class project/test
- Final Exam (50%): project/computer test



Learning resource

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press
- Chollet, F. (2017). *Deep learning with Python*. Manning Publications
- Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola . *Đắm mình trong học sâu*.
Link download [Release Pre-release v0.14.4 · d2l-ai/d2l-vi · GitHub](#) (Nhóm dịch Vũ Hữu Tiệp)
- Nguyễn Thanh Tuấn. Sách Deep learning cơ bản. Link download <https://nttuan8.com/sach-deep-learning-co-ban/>
- Blog: [https:// machinelearningcoban.com](https://machinelearningcoban.com)
- Facebook Page: [https:// www.facebook.com/machinelearningbasicvn/](https://www.facebook.com/machinelearningbasicvn/)
- Facebook Group: [https:// www.facebook.com/ groups/machinelearningcoban/](https://www.facebook.com/groups/machinelearningcoban/)

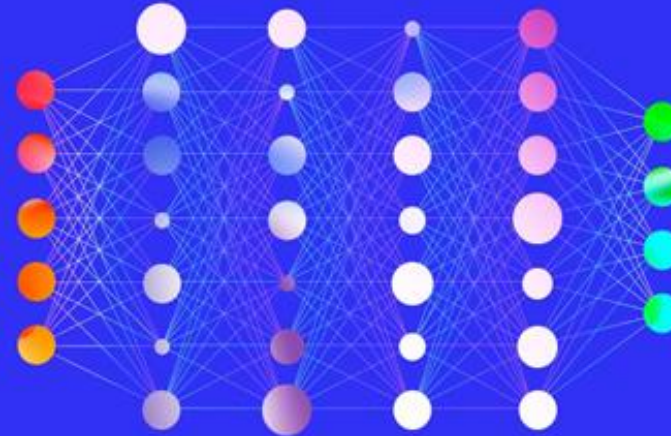
Deep learning

Simple Neural Network



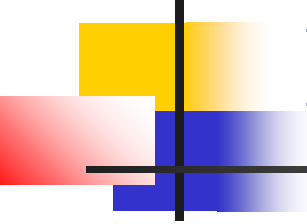
Input Layer

Deep Learning Neural Network

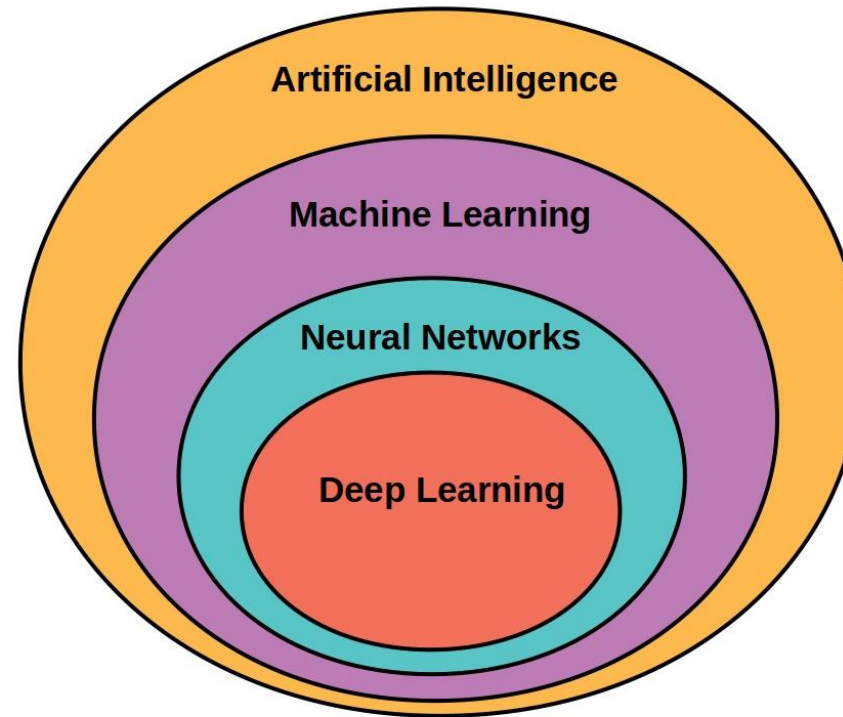


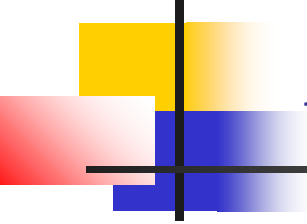
Hidden Layer

Output Layer



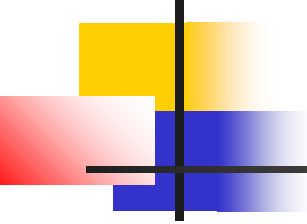
Deep learning vs Machine learning





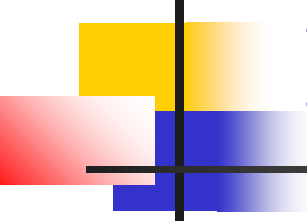
Applications of Deep Learning

- Self-driving cars: Scenario planning, real-time decision-making
- Virtual assistants (e.g., Siri, Alexa): Voice recognition and response
- Image recognition: Auto-tagging, face detection (e.g., Facebook's DeepFace)
- Machine translation: OCR and real-time image/text translation (e.g., Google Translate)
- ...



Open Datasets

- Kaggle: <https://www.kaggle.com/datasets>



Key Programming Libraries

- NumPy: Numerical operations on matrices
- Pandas: Data manipulation and I/O
- Scikit-learn: machine learning methods
- TensorFlow, Pytorch: Deep learning frameworks

Steps to Build a Machine Learning Model

